AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM340
Natural Language Processing

Final Examination
Spring Semester 2025/2026

Date: 3/June/2026

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Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 180 minutes

Instructions:

- Total Marks: **100**
- This exam consists of **three** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

PART 1

Multiple Choice Questions **(20 marks)**

Question 1, (20 marks, 2 mark for each question)
Answer the following MCQ's by selecting **the best** answer for each question. Write your **answer letter** in the answer sheet.

1) Which process involves breaking a stream of text into words, phrases, or symbols?
   A) Normalization
   B) Tokenization
   C) Lemmatization
   D) Stemming
2) In lexical relations, what is the term for words that have the same meaning in some or all contexts?
   A) Antonymy
   B) Hyponymy
   C) Synonymy
   D) Polysemy
3) In an Inverted Index, what is a "Postings List"?
   A) A list of all unique words in the corpus
   B) A list of document IDs where a specific term appears
   C) A list of stop words to be ignored
   D) A matrix of term counts for all documents
4) Why is idf (Inverse Document Frequency) used in ranking?
   A) To give more weight to very common words like "the"
   B) To reduce the weight of rare terms
   C) To calculate the length of a document
   D) To give higher weight to rare terms that are more informative
5) In Part-of-Speech tagging, "Determiners" (a, an, the) belong to which class?
   A) Open Class
   B) Closed Class
   C) Morpheme Class
   D) Variable Class
6) What does a "Dense Vector" (Word Embedding) represent?
   A) A binary presence/absence of a word
   B) A long vector filled mostly with zeros
   C) A short, real-valued vector capturing semantic meaning
   D) The dictionary definition of a word
7) Which activation function is used in the output layer for a binary classification task (e.g., Spam vs. Not Spam)?
   A) ReLU
   B) Tanh
   C) Sigmoid
   D) Linear
8) What is the "Self-Attention" mechanism in Transformers used for?
   A) To reduce the size of the vocabulary
   B) To allow each word to look at every other word in the sequence
   C) To speed up the hardware processing
   D) To replace the word embeddings
9) In a Task-Based Dialogue Agent, what is the goal of "GUS"?
   A) To simulate human emotions
   B) To fill specific slots with information to complete a task
   C) To generate random poems
   D) To act as a search engine

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10) Which metric is used to evaluate how many relevant documents were missed by a search engine?

A) Precision
B) Accuracy
C) Recall
D) F-measure

PART 2

Short Essay Questions **(20 marks)**

Question 2, (20 marks)

Answer the following questions:

a) Define "**Lemmatization**" and explain how it differs from "**Stemming**". (5 marks)
b) Explain why Euclidean Distance is considered a poor measure for document similarity in the Vector Space Model compared to Cosine Similarity. (5 marks)
c) Describe the "**Information Bottleneck**" problem in early Encoder-Decoder architectures. (5 marks)
d) Compare the primary goals of a "**Chatbot**" versus a "**Task-Based Dialogue Agent**". (5 marks)

PART 3

Short problems/questions **(60 marks)**

Question 3, (20 marks)

a) given the following postings lists for three terms:
vector: [3, 7, 12, 18, 25]
semantics: [2, 3, 15, 18, 21, 25]
embeddings: [3, 18, 22, 25, 29, 33]

A user submits the following Boolean query (Q):

(semantics OR embeddings) AND vector

What is the output of applying the merge algorithm to process the query Q, show the required steps and the resulting list of docIDs? (8 marks)

b) Apply **BIO Tagging** to the following sentence identifying ORG (Organization) and LOC (Location) entities: "**Apple is located in Cupertino.**" (6 marks)

c) A search engine returns 10 documents for a query. 4 of them are relevant. The total number of relevant documents in the whole collection is 8. Calculate:
- Precision (3 Marks)
- Recall (3 Marks)

Question 4, (20 marks)

You are given a document collection with N= 1,000,000 documents. A user query is Q = " **neural models**". The raw term frequencies (tf) and document frequencies (df) are provided in the table below (Assume tf in Q = 1 for both terms):

Term	*df*	*tf in D_1*	*tf in D_2*
neural	1,000	10	0
models	10,000	5	10

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Answer the following (**show your calculations for each step**):

a) Calculate the inverse document frequency (idf), and the log-frequency weighting (tf) values for each term? (7 marks)
b) Calculate the ***tf-idf*** for each term, then state decide which term has the highest ***tf-idf*** value in D_1? (6 marks)
c) Calculate the vector length ($|\vec{V}|$) for ***Q, and*** D_2, and ***cos(Q,*** D_1***)***? (7 marks)

Question 5, (20 marks)

A 2-layer feedforward network is used to classify product reviews (y=1 for **Positive Sentiment**, y=0 for **Negative Sentiment**). Both the hidden layer and the output layer use the **sigmoid** activation function.

The features [x1, x2, x3, x4] are defined as:

- x1: Count of the word "excellent"
- x2: Count of the word "poor"
- x3: Set to 1 if the word "not" appears; otherwise, 0
- x4: Set to 1 if the review is shorter than 10 words; otherwise, 0

Analyze the following review:

Quality is poor. I will not recommend it.

The network parameters are:

Layer 1 (Hidden):

$$w_1 = \begin{bmatrix} 0.3 & -0.7 & -1.0 & -0.4 \\ 0.2 & 0.5 & 0.1 & -0.2 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} 0.2 \\ -0.1 \end{bmatrix}$$

Layer 2 (Output):

$$w_2 = [-0.8 \quad 0.6]$$

$$b_2 = 0.5$$

Answer the following (**show your calculations for each step**):

a) Determine the feature vector [x_1, x_2, x_3, x_4] for this review? (4 marks)
b) Calculate the activation vector for the hidden layer? (5 marks)
c) Calculate the activation for the output layer? (4 marks)
d) What is the final classification based on a 0.5 threshold? (2 marks)
e) This specific review is clearly negative, yet the model might produce a specific output. Explain the "**bag-of-words**" limitation in this feature set and suggest how to improve it to handle phrases like "not poor."? (5 marks)

End of questions

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